

Modeling Depression Across Selfish and Cooperative Artificial Societies

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Abstract

We present a theory-driven, mathematically-abstracted model for major depressive disorder (MDD) within the Sugarscape agent-based model. MDD’s measurable diagnostic criteria and prevalence in psychological literature make its modeling straightforward. We demonstrate how disadvantaged agents (such as those diagnosed with MDD) are treated by societies with different attitudes toward cooperation. We compare the outcomes of depressed Sugarscape agents in selfish societies (governed by the principles of ethical egoism) and cooperative societies (governed by the principles of utilitarianism). We further evaluate the impact of the prevalence of MDD in these societies. The resulting 2×2 experimental design identifies how these aspects jointly impact agent interactions and societal outcomes. In particular, agents with MDD in cooperative societies congregate in groups up to 150% in size and have up to 233% greater environmental movement opportunity compared to depressed agents in selfish societies. Cooperative non-depressed agents are willing to sacrifice significant potential resource reward for their actions to promote the welfare of depressed agents, while selfish societies completely eliminate the possibility of self-sacrifice.

Code available at: <https://github.com/digital-terraria-lab/sugarscape>

Data available at: <https://github.com/digital-terraria-lab/datasets>

Introduction

Agent-based modeling (ABM) is a computational approach for examining the interplay of individual agent behaviors within complex, multi-agent systems. By simulating autonomous agents with distinct characteristics, we can explore emergent behavior that arise from agent interactions. ABM aids in investigating phenomena in the social sciences that are impractical to investigate in real life. Limitations of the timescale of a research project, population

size under study, and potential methodological issues in self-reporting are sidestepped by ABM with the compromise that the model cannot capture reality with exact fidelity. We present a computational model for major depressive disorder (MDD) within the *Sugarscape* agent-based model.

In previous work (Kremer-Herman et al., 2024; Kremer-Herman and Milkowski, 2025), we have demonstrated the influence of agent behavior guided by particular ethical lenses on societal success. These lenses are divided between those that prioritize individual, selfish needs and desires (such as ethical egoism) and those that prioritize cooperative, community welfare (such as utilitarianism). Our prior results show cooperative ethical lenses promote far more successful societies than selfish ones across a number of metrics. For systemically-disadvantaged agents, namely those diagnosed with MDD, we present their impact in artificial societies governed by selfish or cooperative lenses.

We use Sugarscape to demonstrate the effect MDD-presenting agents have on society and how societies with non-depressed agents adhering to differing ethical lenses treat depressed agents. We show that there is an effective upper limit to the prevalence of MDD (roughly 50%) in these artificial societies before they experience unavoidable collapse and that cooperative societies are far less susceptible to collapse due to the propensity of agents working together to overcome harsh circumstances. We further show that depressed agents in cooperative societies congregate in groups up to 150% in size and have up to 233% greater environmental movement opportunity compared to depressed agents in selfish societies. Non-depressed agents in cooperative societies are willing to forgo significant potential resource gains for their actions, instead promoting the welfare of depressed agents, while selfish societies obviate the possibility of self-sacrifice.

Sugarscape

Agent-based modeling has long been used to study complex social systems where macro-scale behavior emerges from locally acting agents interacting with one another. Foundational work in social simulation and artificial life

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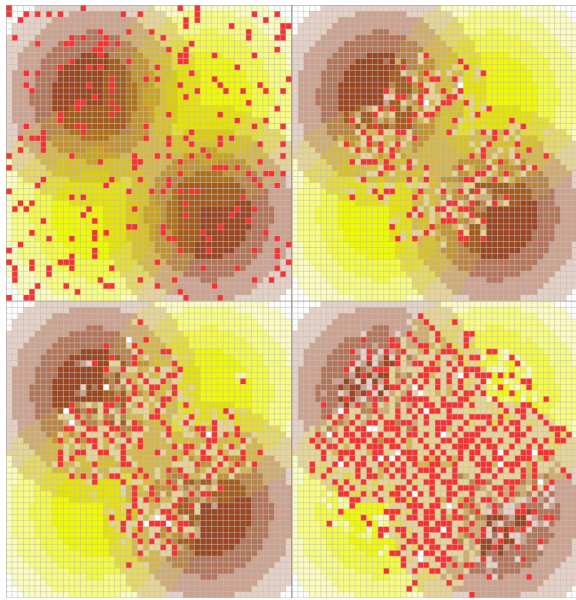


Figure 1: A Sugarscape Society over Time

demonstrate that simple, individual behaviors can potentially give rise to unexpected but rich collective dynamics (Langton, 1995). The Sugarscape ABM was originally presented in *Growing Artificial Societies* (Epstein and Axtell, 1996). Subsequent work has extended Sugarscape to explore more contemporary research questions (Oremland and Laubenbacher, 2014; Pan, 2011; Park et al., 2024). This includes model validation, domain adaptations, and structural modernization (Collins et al., 2024). Recent implementations emphasize transparency, extensibility, and reproducibility to address limitations or discrepancies of earlier versions (Kremer-Herman and Gupta, 2023).

Previous work in artificial life reflects motivations similar to our own. Recent studies highlight the utility of simulated agent societies for exploring human social behavior (Lim and Bentley, 2023; Matsumura et al., 2024), including investigations of socially desirable agent behaviors and emergent societal outcomes shaped by agent values (Matsumura et al., 2024; Merritt et al., 2024). Research on the rapid evolution of neural cognition (Bohm et al., 2022) illustrated the benefit of agent-based systems and computational proxies, specifically their speed, manipulability, and transparency. Our usage of ethical lenses within Sugarscape to examine societal and behavioral outcomes aligns with this field of research.

Our inclusion of MDD finds precedent in work using agent-based disease models to examine susceptible agents and behavioral changes in surrounding agents (Gostoli and Silverman, 2023). Although we do not focus on Sugarscape’s disease capabilities, the same logic applies to the inclusion of MDD. Depressed agents are more susceptible to the environment, and we examine how this alters the behavior of other agents. Furthermore, systems-based mod-

eling has been proposed as necessary in population health research to inform policy decisions (Silverman, 2018). Our analysis of neurodiversity exemplifies this approach, offering insights into MDD treatment and intervention otherwise constrained by traditional approaches.

Sugarscape agents navigate a grid world proliferated with two resources necessary for their survival: *sugar* and *spice*. If an agent is unable to consume enough resources, it dies. Agents interact with each other through combat, cultural propagation, disease spread, friendship, lending, trade, and reproduction. These interactions drive agent *happiness*, representing economic, social, and physical prosperity. Agents may move to a new cell each simulation timestep, selecting the cell in their movement range that yields the highest resource gain (potentially killing another agent and taking a portion of their resources, depending on their *aggression*).

Figure 1 shows four frames of a single short simulation run of Sugarscape without any depressed agents. The top left shows an initial random placement of agents (in red) across the environment with sugar (yellow), spice (brown), and cells with both (tan) spread across the landscape. Greater saturation indicates more resources. The top right displays the simulation at timestep 75 where most agents begin to coalesce in the resource-rich center of the map while others have died due to selection, starvation, or competition. The bottom left frame shows timestep 125 where a stable society can be seen in its beginning stages. Finally, at timestep 200, the society has grown beyond the resource-rich center and can sustain itself toward the edges of the environment.

Agents belong to *tribes* represented by a bitvector of *tags* which are altered when coming into contact with other agents, either reinforcing current tribal membership or converting them. Diseases are contracted from neighboring sick agents and are recovered from over time. Agent tags are also used to identify *friends*, where an agent’s best friend is one they have previously interacted with and has the smallest Hamming distance in their tags. Trading and lending allow agents to obtain resources they cannot access through cell movements alone. Reproduction produces offspring that inherit traits and a stockpile of sugar and spice from the parents. Full details of these features can be found in prior work (Kremer-Herman and Gupta, 2023).

In previous work (Kremer-Herman et al., 2024), we devised ethically-oriented agent goals and preferences beyond simple survival instinct. Particularly, we translated utilitarian and ethical egoist lenses to guide agent utility maximization. For both lenses, utility is scored beyond immediate resource gain (the default in Sugarscape). Factors like how long the gained resources will last the agent and how likely future resource gains are to arise provide a more comprehensive measure of improvement resulting from a movement choice. For utilitarians, this is a democratic process where the moving agent asks its neighbors how they would be impacted by each potential movement. Any move which pro-

motes resource gain for the entire neighborhood is permissible. For egoists, they use this comprehensive utility measure to prioritize benefits only for themselves.

Adjusting these agent features allows us to model MDD. In particular, MDD can present as overeating (increased agent metabolism), irritability (increased aggression factor), lethargy (reduced movement range), and apathy (decreased happiness gains). Finally, agents have a pre-configured maximum number of friends in their social network. Social withdrawal is modeled by decreasing this maximum. MDD is an endemic part of Sugarscape societies. The prevalence of MDD is configurable as a fixed percentage of the population, which is preserved through agent reproduction. Though a simplification, we do not model MDD to present symptoms later in life or in response to stressful situations.

Cultural Modeling and Machine Ethics

The social constructs of individualist and collectivist cultures are central to cross-cultural psychology and sociocultural theory, providing a valuable lens through which to interpret agent behaviors and societal dynamics. Individualism emphasizes personal autonomy, self-interest, and the pursuit of individual goals, often at the expense of group cohesion or welfare (Hofstede, 2001; Triandis, 1995). In contrast, collectivism emphasizes group harmony, interdependence, and the success of the community equating to success for the individual (Hofstede, 2001; Triandis, 1995).

Previous research has found that lifetime prevalence of depression is higher in countries commonly identified as individualist, such as the United States and Canada, than in those commonly identified as collectivist, like China and Japan (Kessler and Bromet, 2013; Ferrari et al., 2013). This pattern aligns with our findings regarding selfishness and cooperation in ABM (Kremer-Herman et al., 2024). Ethical egoism and utilitarianism share similar qualities to individualism and collectivism, respectively.

We leverage our common understanding of Sugarscape to similarly investigate societies where some agents experience MDD. We caution that we are not implying that our work somehow is prescriptive for any particular locale or society in human life. Rather, to the extent that modeling captures key aspects of a society, ABM facilitates the analysis of how cultural predispositions interact with neurodiversity to produce complex emergent social systems.

Machine ethics primarily concerns the design of autonomous agents capable of making decisions informed by moral considerations Anderson and Anderson (2011). This particular line of research motivates the need for digital environments where moral algorithms can be observed, compared, and stress tested over an arbitrary amount of time with configurable degrees of complexity Kremer-Herman et al. (2024). In this case, we model MDD and stress test how two moral algorithms treat these disadvantaged agents.

Modeling Neurodiverse Symptomatology

Psychological research has faced long-standing challenges such as reliance on self-report measures (Nichols and Maner, 2008; Kihlstrom, 2002), isolating disorders from frequent co-diagnoses (Steffen et al., 2020; Kaiser et al., 2021), and monitoring longitudinal effects. Additionally, individual social factors, such as race and socioeconomic status, often further blur the interpretation of the phenomena in question. Using ABMs to model psychological phenomena provides a method to isolate relevant factors for interpretation and quantify longitudinal effects at a societal scale.

We have chosen to illustrate the use of this approach through a theory-driven abstraction of major depressive disorder which is an episodic and irregular psychological disorder with a high prevalence rate that increased significantly from 2015-2020 (Goodwin et al., 2022). Despite more than 208 available self-report measures (Santor et al., 2006) and substantial scientific interest, MDD research remains constrained by the influence of individual social factors and the presence of symptoms that overlap with other disorders, such as generalized anxiety disorder and post-traumatic stress disorder (Klein Breteler et al., 2021). Since confounding factors cannot be fully controlled in human subjects research, an ABM provides a potentially powerful framework to better understand the effects of innate neurodiversity, in this case MDD.

Table 1: *Final Sugarscape Parameters*

Fixed Parameters	Value
Initial agents	250
Maximum timesteps	2,000
Random seeds	200
Varied Parameters	Value
MDD prevalence	0%–50% in 5% steps
Societal ethical lens	Selfish / Cooperative
Agent type	Control / Depressed
MDD Agent Manipulations	Value
Happiness	$\times 57.63\%$
Max friends	-36.67%
Metabolism	$+54.4\%$
Movement	$-19.6\%, -37.5\%$
Aggression	$+14.5\%$

Methodology

Table 1 summarizes the model settings and parameters central to our experimentation. We include fixed simulation setup parameters for all configurations, varied parameters regarding MDD and ethical lenses, and the effects of MDD on agents. Selfish societies, reflective of ethical egoism, operate under the principle that each agent only prioritizes their

own survival. In the simulation, each agent always chooses the cell that offers the most resources to ensure their survival. Cooperative societies, reflective of utilitarianism, operate under the principle that agents consider the resource needs of neighboring agents. Although personal survival remains the priority, an agent can forgo the cell with the best resources in favor of one that will maintain their livelihood while also allowing neighboring agents to obtain required resources. Further details regarding the translation of the selfish and cooperative ethical lenses are provided in prior work (Kremer-Herman et al., 2024).

We varied two configurable parameters in Sugarscape for our experimentation: the population prevalence of depression and the societal ethical lens. Initially, 21 depression prevalence rates were considered, ranging from 0% to 100% in 5% intervals, and two ethical lenses, selfish and cooperative, were used. Pilot testing found that societies exceeding a prevalence rate of 50% failed to survive. As such, we narrowed the experimental design to 11 depression prevalence rates (0% to 50% in 5% increments). These conditions were fully crossed, creating 22 unique simulation conditions. Our testbed generated 200 seeds, which create 200 randomized initial states for each condition.

Simulating Depression

Since depression prevalence does not remain constant in real populations (Moreno-Agostino et al., 2021; Shorey et al., 2022; Ormel et al., 2022; Kim et al., 2020), we explore emergent differences across varied prevalence rates. A number of the starting Sugarscape agents, determined by the configured prevalence rate, exhibit MDD symptoms. When agents reproduce, the prevalence rate is used as a percentage chance that any offspring will exhibit MDD symptoms.

The diagnostic criteria of MDD, as published in the Diagnostic and Statistical Manual of Mental Disorders (DSM-5), states a minimum of five of the nine possible symptoms of MDD are required for diagnosis (APA, 2022). All depressed agents in Sugarscape manifest six symptoms: depressed mood, diminished interest, weight changes, sleep changes, fatigue, and psychomotor changes. We exclude the remaining symptoms (feelings of unworthiness, diminished concentration, and recurrent thoughts of mortality) as they lack behavioral indicators and cannot be accurately modeled (APA, 2022). This design decision trades generalizability for computational feasibility. To account for some symptom heterogeneity, we manipulate heterogeneous agent parameters (such as metabolism and movement range) via a fixed percentage adjustment.

Depressed Mood. Depressed mood is a commonly diagnosed symptom among individuals with MDD (Fried and Nesse, 2015; Rottenberg et al., 2005) that is also used as a risk factor for future diagnosis (Cuijpers and Smit, 2004). A previous study (Koivumaa-Honkanen et al., 2004) showed

that those with moderate to severe depression, indicative of MDD, reported lower life satisfaction scores ($\tilde{x} = 19$) compared to those with no to mild depression ($\tilde{x} = 10.5$). Accordingly, depressed mood is modeled as a reduction in agent happiness such that agents only retain 57.63% of their baseline happiness gains from their activities.

Diminished Interest. *Anhedonia*, a loss of interest in previously enjoyed activities, can lead to social withdrawal (Dold et al., 2021). It is experienced by 60.71% of MDD patients, causes further symptom vulnerability (APA, 2022; Loonen and Ivanova, 2016; Dold et al., 2021; Kupferberg et al., 2016) and produces exaggerated reactions to negative events with dampened reactions to positive events (Uwajeh et al., 2024; Zlotnick et al., 2000). Because anhedonia both exaggerates negative interactions and dampens positive ones, directly comparing MDD to non-depressed conditions would conflate these two effects into a single parameter. Instead, we estimated the effect of anhedonia by comparing negative interaction rates in MDD to positive interaction rates in *dysthymia*, which is a chronic, mild depression that excludes anhedonia as a diagnostic criterion (Zlotnick et al., 2000; National Institute of Mental Health, nd; APA, 2022). Since dysthymia does not involve anhedonia, positive interaction rates remain unaffected (Harvard Health Publishing, 2014), providing a clean baseline for comparison. The resulting anhedonia-driven social withdrawal percentage indicates a 36.67% elevation in negative interactions relative to this baseline, which we used to penalize an agent's maximum friend count as the closest matching metric to social withdrawal in Sugarscape.

Weight Changes. Weight and MDD share a bidirectional relationship, with overlapping mechanisms linking MDD to appetite dysregulation (Mills et al., 2018; Stensel, 2010). The DSM-5 classifies both weight gain and loss as depressive symptoms, defined as a 5% body weight change within one month (APA, 2022). To parameterize this, we calculated a baseline daily caloric intake (Cleveland Clinic, 2025) and a deficit intake sufficient to reach the DSM-5 weight-loss threshold (UF Health, 2022). Averaging published recommendations across genders yields a baseline of 2,250 calories per day. For the deficit, we averaged a very low-calorie diet (VLCD; 800 kcal) with the midpoint across genders of a low-calorie diet (LCD; 1,000–1,600 kcal, $M = 1,250$), producing a weight-loss intake of 1,025 calories per day. Agent metabolisms for sugar and spice are increased by 54.4%, derived from the percentage decrease between the weight-loss and baseline intake values. This formulation captures both ends of appetite dysregulation: undereating is represented by a harder-to-reach metabolic requirement, while overeating is represented by greater resource consumption. A higher metabolism also represents a hindrance in resource competition, reflecting the negative valence of the symptom.

Changes in Sleep and Fatigue. Fatigue is present in 95% of MDD patients (Ghanean et al., 2018). The residual (persistent) nature of fatigue causes feelings of exhaustion and affects individuals’ capacity for work and activities (Fava et al., 2014). Fatigue results from both *insomnia* and *hypersomnia* (Zhao et al., 2021), making it well-suited to capture both independent fatigue and changes in sleep. A previous study indicates that fatigue is present in 4.7% of individuals without depression and in 24.3% of individuals with severe depression (Robinson et al., 2015). We deduct non-depressed fatigue from depression-attributable fatigue, creating a 19.6% penalty for agent movement.

Psychomotor Changes. *Psychomotor agitation*, which is increased movement or mood, and its rough opposite *retardation*, which refers to slowed cognition and movement, are experienced by up to 70% of individuals with MDD (Sobin, 1997; Wüthrich et al., 2024). They offer high discriminative validity for differentiating depression from confounds (Wüthrich et al., 2024). For agitation, aggressive outburst prevalence was 58.7% in individuals with MDD compared to 44.2% in those without (Liu and Cole, 2021). This difference is applied as a 14.5% increase to base agent aggression. For retardation, productivity impairment serves as a proxy for psychomotor slowing given its direct impact on job performance. One study reported impairments of 28% for mild and 47% for severe depression (Beck et al., 2011). Since we model no distinction between mildly and severely depressed agents, the reported percentages were averaged, resulting in a 37.5% productivity impairment penalizing agent movement range.

Agent Behavior Outcome Measures

We study emergent behavior in agent interactions through neighborhood-level and decision-level outcomes which rely on cells in an agent’s movement range. These cells form a ranked list of choices based primarily on resources required for survival. Reachable cells occupied by other agents form an agent’s neighborhood. Choosing a cell that is not the best-ranked option defines a sub-optimal decision.

We extrapolate societal structure from neighborhood composition among agents to identify emergent clustering or segregation preferences. To examine the intersection of experimental conditions, we compared outcomes for depressed agents across selfish and cooperative societies. Sub-optimal decisions are observed in cooperative societies only, where agents may sacrifice their best cell (which has the most resources) for the overall benefit of their neighborhood. Since selfish agents always choose the best-ranked cell, this metric is not applicable for those societies. Our experiments suggest that the ethical preferences of a society have an outsized effect on neurodiverse agents relative to control agents.

Results

We examine societal-level outcomes and agent behavior to provide a comprehensive understanding of society dynamics. All reported results are aggregated across seeds for each unique pairing of societal ethical lens and depression prevalence. We begin at the societal level, examining societal longevity to determine how societies are broadly altered by different agent types and societal ethics. We then shift to agent behavior, measured by neighborhood composition, social interactions among agents, and the occurrence of sub-optimal decisions. This reflects the extent to which agents sacrifice personal gain for the neighborhood’s benefit.

Table 2: *Societal Survival*

Behavior	Depressed	Extinct	Stable
Cooperative	0%	2	198
Selfish	0%	85	115
Cooperative	25%	28	172
Selfish	25%	140	60
Cooperative	50%	178	22
Selfish	50%	192	8
Cooperative	75%	200	0
Selfish	75%	200	0
Cooperative	100%	200	0
Selfish	100%	200	0

Societal Survival

A clear metric of societal success is whether it survives over time. Table 2 presents societal survival outcomes across quartile benchmarks of depression prevalence, stratified by agent behavior type. Societies were classified by population after 2,000 timesteps: *extinct* indicates societal collapse and *stable* indicates a surviving agent population. Previous work (Kremer-Herman et al., 2024) indicates this runtime is sufficient for reaching a steady societal state.

At 0% depression, cooperative societies outperform selfish by minimizing society extinction. This cooperative advantage becomes critical at 25% depression, where selfish societies collapse dramatically, and cooperative societies remain relatively strong. At 50% and beyond, most societies regardless of ethical framework reach a steady state (extinction or stability), tending towards extinction.

Impact of Neighborhood Size

To gauge how behavior from ethical lenses results in differing treatment of neurodiverse agents, we investigate differences in neighborhood sizes. Recall that an agent’s neighborhood is the set of other agents in its movement range. Relative results of cooperative and selfish societies were examined across mean neighborhood size, depressed neigh-

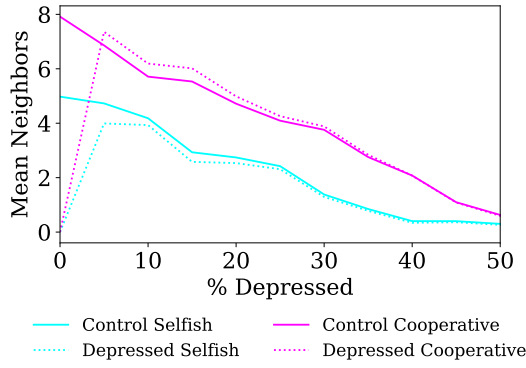


Figure 2: Mean Neighbor Composition. Average number of agent neighbors, aggregated across seeds for intervals of population depression percentages.

bor composition, and control neighbor composition. For these outcomes, levels decrease as depression prevalence increases, which is an artifact of increased societal extinction.

Figure 2 displays the average neighborhood size by agent type across depression prevalence rates. Cooperative societies maintain larger neighborhoods compared to selfish ones. There is a non-uniform interaction between agent type and societal structure, where cooperative depressed agents have the largest neighborhoods and selfish depressed agents have the smallest. This indicates neighborhood size is dependent upon agent type beyond the size increase resulting from cooperation. For depressed agents, cooperative behavior and larger social networks provide something desirable.

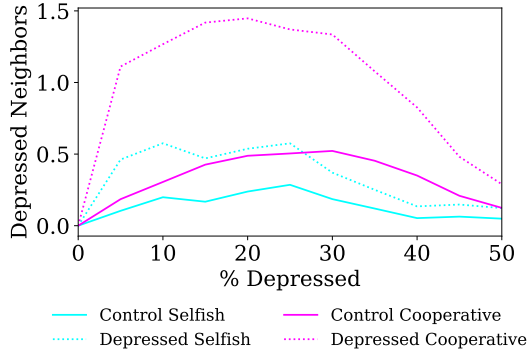


Figure 3: Mean Depressed Neighbor Composition. Average number of depressed agent neighbors aggregated across seeds for intervals of population depression percentages.

The distinction between agent type is enhanced in Figure 3, showing the average number of depressed agents in a neighborhood. Depressed agents tend to cluster with each other, as their control counterparts maintain fewer depressed neighbors. Cooperative depressed agents display a unique clustering pattern arising from cooperative decisionmaking.

Cooperative behavior enables these agents to remain in close social groups without compromising their survival.

When counting the number of control agent neighbors in Figure 4, cooperative depressed agent clustering explains why they have the largest neighborhood size in Figure 2. The mean number of control neighbor agents follows a similar society-dependent pattern seen in total neighborhood size. This is where we see the deviation of neighborhood choice. Cooperative depressed agents cultivate the largest neighborhoods by making an effort to remain with like agents. This is new, emergent behavior.

As shown in Figure 3, holding the acting agent's type constant, cooperative agents have more depressed agents in their neighborhoods than selfish agents. This is a result of cooperation since these agents are less concerned with resource competition, allowing the formation of larger neighborhoods. Selfish societies cannot have this same grouping behavior as they never work in unison. Cooperation provides larger communities with greater support for all agents, but even more so for neurodiverse individuals, by reducing the pressures of competition that can cause agents to disperse.

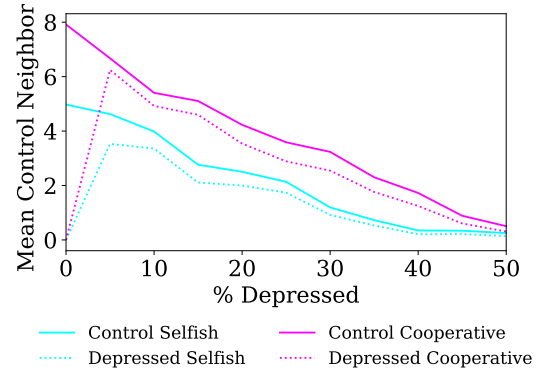


Figure 4: Mean Control Neighbor Composition. Average number of control agent neighbors aggregated across seeds for intervals of population depression percentages.

The Benefit of Suboptimal Decisions

A key distinction between selfishness and cooperation is a willingness to forgo one's own gain for the wellbeing of others. We investigate the degree to which cooperative agents make sacrifices for others, particularly control agents giving up opportunities to benefit depressed agents. Relative results of cooperative and selfish societies were examined across the magnitude of lost resources, the suboptimality of agent decisions, and the number of cell options available for agent movement. For these figures, levels decrease as depression prevalence increases, which is an artifact of increased societal extinction providing fewer possible agent interactions.

When studying suboptimal decisions, the existence of large neighborhoods among agents is particularly telling.

A suboptimal decision is made when an agent eschews the highest value cell (in terms of raw resource gain) in favor of a more cooperative choice. This suboptimal choice creates additional degrees of freedom for agents who will act later, allowing them to select those high-value cells. Such events steadily decline as the depressed population percentage climbs; with a lower population, there is less urgency to make a suboptimal decision. The existence of suboptimal decisions is tightly coupled with resource availability, procurement, and overall agent survival. This showcases the importance of agent interactions on societal survival, especially when societies are composed of multiple agent types.

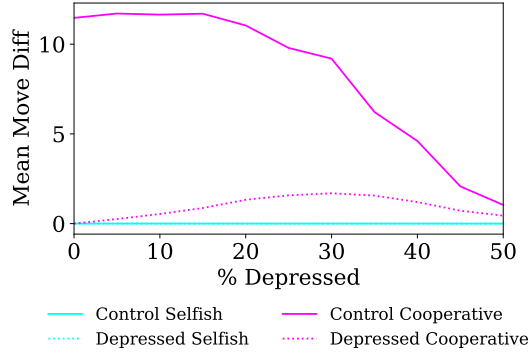


Figure 5: Mean Move Difference. Difference in obtained and maximum possible resources aggregated across seeds for intervals of population depression percentages.

Figure 5 shows the mean magnitude of resources lost when an agent moves to a suboptimal cell. Selfish agents have a difference of zero, since they always, greedily, take their optimal movement choice. For cooperative societies, we find that control agents lose a greater amount of resources than depressed agents when making a suboptimal decision. This outsized loss in resources is, in part, because these agents opt to not murder other agents (depressed or otherwise) in the pursuit of raw resource gain. In essence, control agents are leaving open the most attractive cells for those agents whose need may be greater.

Depressed agents require more resources to survive each timestep because of the metabolism penalty, effectively capping the resources they can lose without a decision resulting in death. Their movement penalty further restricts accessible cells, potentially reducing the gap between best and worst resource choices. Control agents do not have these same constraints, providing more options that are less deadly. Additionally, agents store and carry resources. Although this is not directly penalized in depressed agents, we conclude that their higher metabolisms result in fewer leftover resources. Control agents are able to subsist on their reserves when bringing in substantially fewer resources in a timestep.

For every suboptimal decision, we recorded the position of the chosen cell in an agent’s ranked list. Figure 6 depicts

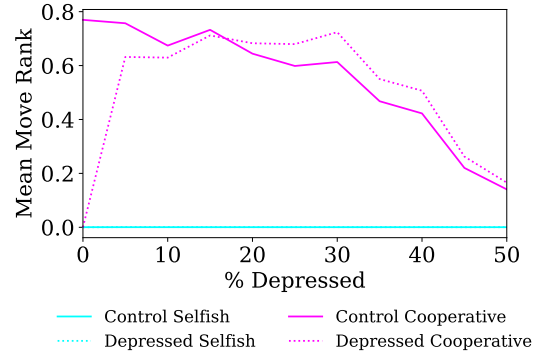


Figure 6: Mean Move Rank. Difference in rank positions between the chosen and best cell, aggregated across seeds for intervals of population depression percentages.

the rank difference from the best move as another measure of suboptimality. Since selfish agents cannot make suboptimal decisions, their rank difference is always zero.

In Figure 6, we see that depressed agents sometimes make suboptimal choices, which seems counterintuitive since they tend to have fewer resources than control agents. This is a result of the cooperative decisionmaking at play. Like control agents, depressed agents also make choices based on their neighborhood’s wellbeing. In the observed emergent behavior of depressed agent clustering, there are fewer available cells for movement. The magnitude of the difference in resources lost by suboptimal decisions for the neighborhood is not significantly different. In other words, the net resource gain of the neighborhood changes little due to their locality.

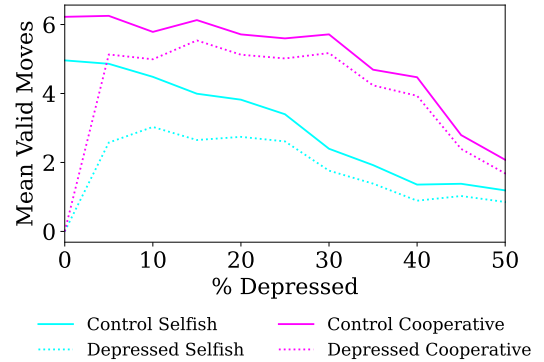


Figure 7: Mean Valid Moves. Average number of possible cell choices per agent, aggregated across seeds for intervals of population depression percentages.

Depressed agent clustering influences their cell choices by prioritizing proximally close cells for the sake of remaining with the group. When they make suboptimal decisions, they are less willing to suffer large differences in resource magnitude, whereas control agents are more capable of with-

standing a loss of many resources. Depressed agents making choices to remain grouped beyond maintaining neighborhood survival is a notable emergent behavior since there is no element of ethics-derived decisionmaking that prioritizes remaining with the neighborhood.

Figure 7 shows the mean number of possible valid moves available to an agent. Depressed agents have relatively fewer move options than control agents due to their movement penalty. Since all selfish agents choose their best cell, cell options for other agents decrease as agents take their turn. This competitive nature disproportionately affects depressed agents, since they have fewer cell options to begin with.

In cooperative societies, the gap between valid moves for depressed and control agents narrows. This results from cooperative decisionmaking. Depressed agents have more of their original cell options available as other agents leave these cells available to them. We propose that control agents create these more favorable circumstances for their depressed agent peers. With larger neighborhoods in cooperative societies, control agents can more freely choose cells farther away from the group and leave cell options open for their neighboring depressed agents. When control agents remove themselves from densely populated depressed areas, there are more options for depressed agents to take the path of least resistance while ensuring group survival.

Reproducibility

We used the `v2026.2` release of Sugarscape to produce our results. We provide the JavaScript Object Notation (JSON) configuration files required to reproduce the dataset, which are found in the `modeling-depression.zip` archive. Details concerning the options for Sugarscape and the helper scripts are found in the `README`.

Before execution, unzip and place the dataset configuration files in the repository’s `data` directory. Run `make data` to start data collection. Run `make plots` to generate graphs, so long as the `matplotlib` module is installed. Modifying the script in the `plots` directory is needed to reproduce the graphs presented in this work.

Conclusions

Across our analyses, selfish societies are worse for depressed agents as they have the fewest cell options for survival. Cooperative societies benefit depressed agents by increasing their cell options and minimizing resource loss. Since this behavior arises from cooperative decisionmaking, these societies more effectively mitigate MDD-associated impairment through group support. Furthermore, the positive correlation between neighborhood size and valid moves indicates that societal outcomes rely on the decision process, rather than density-driven resource competition.

Depressed agents congregate in both societies, but it is most pronounced among cooperative depressed agents who collectively make choices to remain together. Control agents

have different behaviors depending on society types; as depression prevalence increases, cooperative agents increase their number of depressed neighbors, while selfish agents do not. This is governed by agent interaction and ethical decisionmaking. Cooperative control agents prioritize making neighborhood-based decisions which as a consequence benefits their depressed peers. Selfish control agents have no consideration for others and do not differentiate between moving near a depressed agent or away from one.

Beyond societal observations, we show a viable approach to modeling neurodiversity, providing a methodology for psychological research unaffected by common constraints. Our theory-driven abstraction enables future investigation into emergent phenomena at the interaction of impairment, societal norms, and social policy. We acknowledge our model is a computational proxy for MDD through a specific subset of diagnostic criteria. As such, our results illustrate system dynamics rather than a full characterization.

When considering beneficial treatment for neurodiverse individuals, our findings indicate that the artificial society’s ethical lens may be as important as individual disorder management. Cooperative behavior can foster networks that mitigate isolation and fundamentally improve quality of life. These priorities create systemic benefit for all individuals through behavior promoting welfare and survival. The ethical effects captured at a societal level likely extend to smaller scales, such as communities and workplaces, where similar benefits of cooperation and social support may emerge.

Future Work

Our chief direction for future work is improving our MDD model by introducing symptom heterogeneity. Since MDD presents differently across sexes, the probability of exhibiting a symptom could be conditioned on agent sex. We could also include other forms of neurodiversity in the model, broadening its validity for behavioral research.

Sugarscape also collects aggregated data about agents such as resource wealth, life expectancy, happiness, and other intensity-driven survival metrics. These data present an opportunity for future work to incorporate these metrics in a more comprehensive evaluation of not only the life of depressed agents but the quality of that life. We can similarly gauge societal success by these metrics, and doing so might provide deeper insight into how different societal structures support or disadvantage agents with MDD.

We intend to further investigate the impact of ethical lenses on societal-level outcomes. Our results show that a cooperative, utilitarian framing of the world leads to sharp differences in the treatment of depressed agents. Other ethical lenses, like Kantian deontology or social contractarianism, encourage cooperation from radically different perspectives. We anticipate there are numerous ethical lenses for which depressed agents are treated with relative compassion compared to the selfish lens offered by ethical egoism.

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